

Submission Worksheet

CLICK TO GRADE

<https://learn.ethereallab.app/assignment/IT114-005-F2024/it114-milestone-3-chatroom-2024-m24/grade/bna24>

Course: IT114-005-F2024
Assignment: [IT114] Milestone 3 Chatroom 2024 M24
Student: Bakr A. (bna24)

Submissions:

Submission Selection

1 Submission [submitted] 11/27/2024 11:40:38 PM

Instructions

^ COLLAPSE ^

Implement the Milestone 3 features from the project’s proposal document:
<https://docs.google.com/document/d/1ONmvEveI97GTFPGfVwwQC96xSsobbSbk56145XizQG4/view>
Make sure you add your ucid/date as code comments where code changes are done All code changes should reach the Milestone3 branch Create a pull request from Milestone3 to main and keep it open until you get the output PDF from this assignment. Gather the evidence of feature completion based on the below tasks. Once finished, get the output PDF and copy/move it to your repository folder on your local machine. Run the necessary git add, commit, and push steps to move it to GitHub Complete the pull request that was opened earlier Upload the same output PDF to Canvas

Branch name: Milestone3

Group

100%

Group: Basic UI
Tasks: 1
Points: 2

^ COLLAPSE ^

Task

100%

Group: Basic UI
Task #1: UI Panels
Weight: ~100%
Points: ~2.00

Details:

All code screenshots must include ucid/date.

App screenshots must have the UCID in the title bar like the lesson gave.

Columns: 1

Sub-Task

Group: Basic UI

Task #1: UI Panels

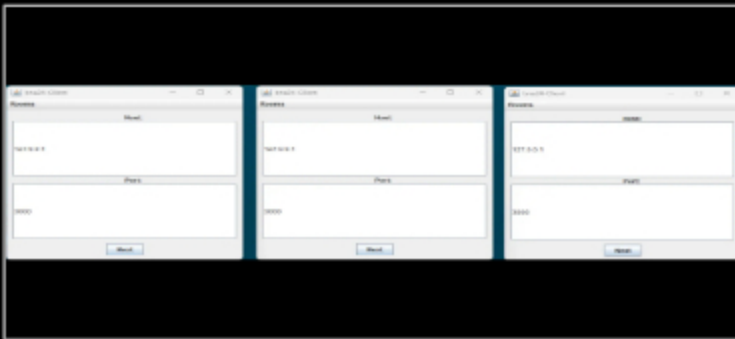
Sub Task #1: Show the ConnectionPanel by running the app (should have host/port)

100%

Task Screenshots

Gallery Style: 2 Columns

4 2 1



ConnectionPanel UI

Caption(s) (required) ✓

Caption Hint: Describe/highlight what's being shown

Sub-Task

Group: Basic UI

Task #1: UI Panels

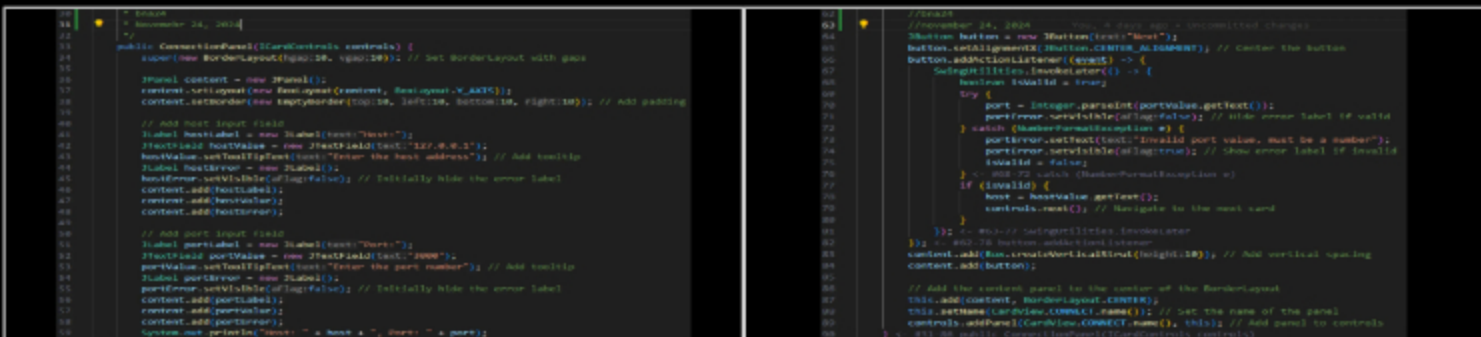
Sub Task #2: Show the code related to the ConnectionPanel

100%

Task Screenshots

Gallery Style: 2 Columns

4 2 1



Host and Port Input fields

next button

Caption(s) (required) ✓
Caption Hint: *Describe/highlight what's being shown*

Task Response Prompt

Briefly explain how it works and how it's used

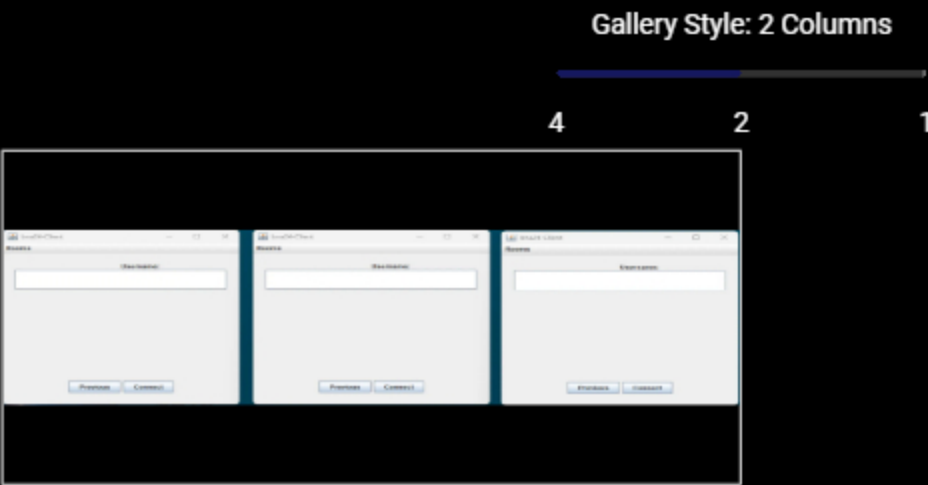
Response:

The ConnectionPanel provides a user interface for entering the host and port required to connect to a server. It includes input fields for the host (default: 127.0.0.1) and port (default: 3000), along with a "Next" button to validate the inputs. If the port is invalid, an error message is displayed; otherwise, the entered values are stored, and the panel transitions to the next screen using the provided ICardControls instance. It's used in a card based layout system.

Sub-Task

Group: Basic UI
Task #1: UI Panels
Sub Task #3: show the UserDetailsPanel by running the app (should have username)

Task Screenshots



UserDetailsPanel

Caption(s) (required) ✓
Caption Hint: *Describe/highlight what's being shown*

Sub-Task

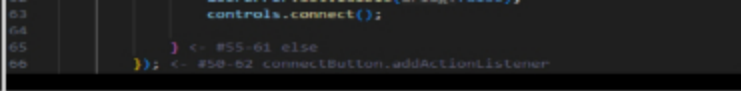
Group: Basic UI
Task #1: UI Panels
Sub Task #4: Show the code related to the UserDetailsPanel

Task Screenshots

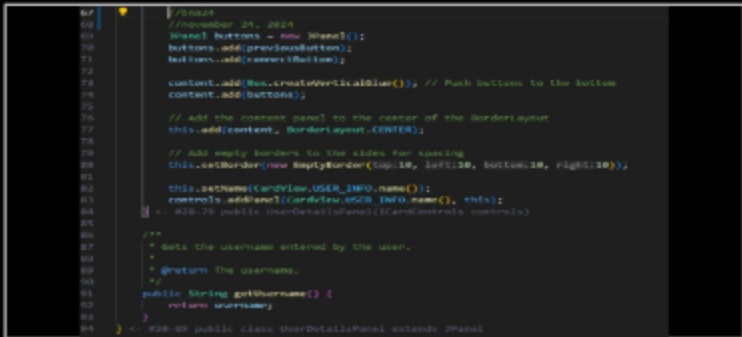




Username Input



Adds previous and Connect Buttons



Cosmetic Adjustments

Caption(s) (required) ✓

Caption Hint: *Describe/highlight what's being shown*

≡ Task Response Prompt

Briefly explain how it works and how it's used

Response:

The UserDetailsPanel features a text field for username input, an error label for validation, and two buttons: "Previous" for navigating back and "Connect" for proceeding. The "Connect" button validates the input to ensure the username is not empty, stores it, and triggers the controls.connect() method to move to the next step. It integrates into an application by registering with a card control interface and captures user input for subsequent operations like connecting to a server.

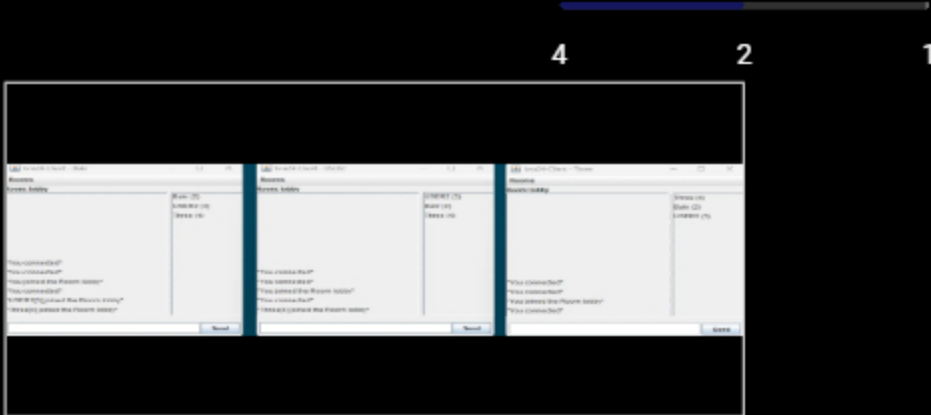
Sub-Task

100%

Group: Basic UI
Task #1: UI Panels
Sub Task #5: Show the ChatPanel (there should be at least 3 users present and some example messages)

🖼 Task Screenshots

Gallery Style: 2 Columns



3 users in the ChatPanel

Caption(s) (required) ✓

Caption Hint: *Describe/highlight what's being shown*

Sub-Task

100%

Group: Basic UI
Task #1: UI Panels

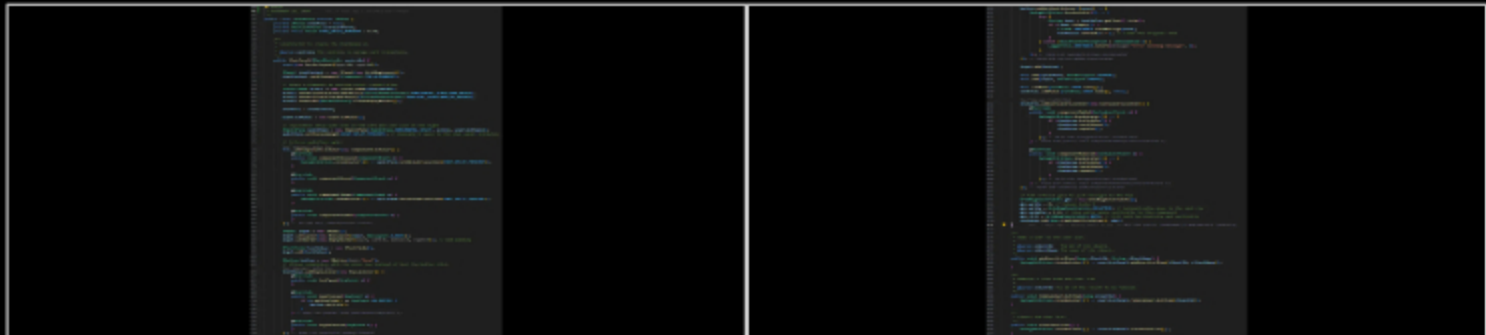
Task Screenshots

Gallery Style: 2 Columns

4

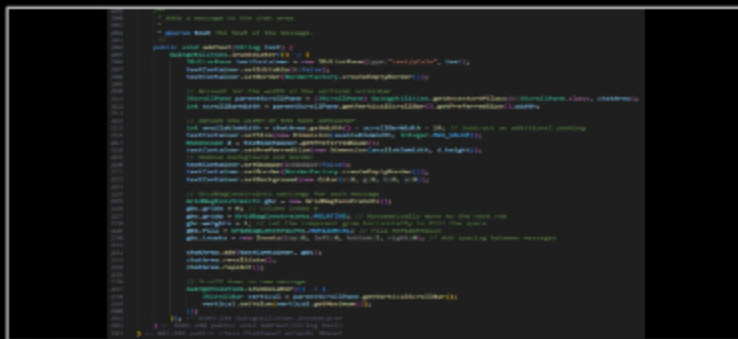
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1



ChatPanel #1

ChatPanel #2



ChatPanel #3

Caption(s) (required) ✓

Caption Hint: *Describe/highlight what's being shown*

Task Response Prompt

Briefly explain how it works and how it's used (note the important parts of the ChatPanel)

Response:

The ChatPanel is the main UI for managing chat interactions. It uses a BorderLayout to organize components, with the chat messages displayed in a JScrollPane containing a JPanel (chatArea) using GridBagLayout. Messages are added via the addText method, which formats and dynamically inserts them into the chatArea. A JSplitPane divides the chat view and the UserListPanel, which manages connected users through methods like addUserListItem and removeUserListItem. The input field and "Send" button at the bottom allow users to send messages, triggering Client.INSTANCE.sendMessage(). Key bindings, like pressing Enter, also invoke the send action.

End of Task 1

End of Group: Basic UI
Task Status: 1/1

Group

Group: Build-up
Tasks: 2

100%

Points: 3

^ COLLAPSE ^

Task

100%

Group: Build-up

Task #1: Results of /flip and /roll appear in a different format than regular chat text

Weight: ~50%

Points: ~1.50

^ COLLAPSE ^

Details:

All code screenshots must include ucid/date.

App screenshots must have the UCID in the title bar like the lesson gave.



Columns: 4

Sub-Task

100%

Group:

Build-up

Task #1:

Results of

/flip and

/roll

appear in a

Sub-Task

100%

Group:

Build-up

Task #1:

Results of

/flip and

/roll

appear in a



Task

Screenshots

Gallery Style: 2 Columns

4 2 1



Different
Formats for
Flip and Roll

Caption(s) (required) ✓

Caption Hint:

Describe/highlight what's
being shown



Task

Screenshots

Gallery Style: 2 Columns

4 2 1



handleRoll and
handleFlip

Caption(s) (required) ✓

Caption Hint:

Describe/highlight what's
being shown

≡ Task

Response

Prompt

Explain what you did and how

it works

Response:

/roll and /flip commands return with their results colored with specific color tags. In handleRoll, the roll result uses blue color, #b...b#, and in handleFlip, 'heads' or 'tails' in green, #g...g#. This formatting is applied server side through the formatText function, so that the results are formatted differently from regular messages when they present themselves in the chat.

End of Task 1

Task



Group: Build-up

Task #2: Text Formatting appears correctly on the UI

Weight: ~50%

Points: ~1.50

^ COLLAPSE ^

Details:

All code screenshots must include ucid/date.

App screenshots must have the UCID in the title bar like the lesson gave.



Columns: 1

Sub-Task



Group: Build-up

Task #2: Text Formatting appears correctly on the UI

Sub Task #1: Show examples of bold, italic, underline, each color implemented and a combination of bold, italic, underline, and one color in the same message

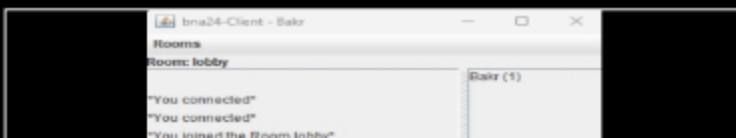
Task Screenshots

Gallery Style: 2 Columns

4

2

1




```

You connected
Bakr[1] <b>This is bold</b>
Bakr[1] <i>This is italic</i>
Bakr[1] <u>This is underlined</u>
Bakr[1] <red>This is red text</red>
Bakr[1] <green>This is green text</green>
Bakr[1] <blue>This is blue text</blue>
Bakr[1] <b><u><i><red>This is bold, italic, and
underlined, and red</red></i></u></b>

```

Send

UI formatting

Caption(s) (required) ✓

Caption Hint: *Describe/highlight what's being shown*

Sub-Task

Group: Build-up

100%

Task #2: Text Formatting appears correctly on the UI

Sub Task #2: Show the code changes necessary to get this to work

Task Screenshots

Gallery Style: 2 Columns

4

2

1

```

314 // Text Formatting
315 // bma24
316 // november 11, 2024
317 private String formatText(String message) {
318     String boldPattern = "\\b(.*?)\\b";
319     String italicPattern = "\\i(.*?)\\i";
320     String underlinePattern = "\\u(.*?)\\u";
321     String redPattern = "\\r(.*?)\\r";
322     String greenPattern = "\\g(.*?)\\g";
323     String bluePattern = "\\b(.*?)\\b";
324
325     //replaces with HTML
326     message = message.replaceAll(boldPattern, replacement: "<b>${1}</b>");
327     message = message.replaceAll(italicPattern, replacement: "<i>${1}</i>");
328     message = message.replaceAll(underlinePattern, replacement: "<u>${1}</u>");
329     message = message.replaceAll(redPattern, replacement: "<red>${1}</red>");
330     message = message.replaceAll(greenPattern, replacement: "<green>${1}</green>");
331     message = message.replaceAll(bluePattern, replacement: "<blue>${1}</blue>");
332
333     return message;
334 } // 317-334 private String formatText(String message)

```

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```

formatText

addText

Caption(s) (required) ✓

Caption Hint: *Describe/highlight what's being shown*

Task Response Prompt

Briefly explain what was necessary and how it works

Response:

To format text, the formatText method in the Room class was modified to do bold, italic, underline and colored text in regular expressions. The patterns like `text` was replaced with HTML tags such as `<b>` for bold. By doing this the formatting will be the same throughout all the messages that get processed on the server. The ChatPanel was then updated on the client side to write messages using text/html rather than plain text so it could render with formatting in the chat interface.

End of Task 2

End of Group: Build-up

Task Status: 2/2

Group

Group: New Features

Tasks: 2

100%

Points: 4

^ COLLAPSE ^

Task

100%

Group: New Features

Task #1: Private messages via @username

Weight: ~50%

Points: ~2.00

^ COLLAPSE ^

Details:

All code screenshots must include ucid/date.

App screenshots must have the UCID in the title bar like the lesson gave.



Columns: 1

Sub-Task

100%

Group: New Features

Task #1: Private messages via @username

Sub Task #1: Show a few examples across different clients (there should be at least 3 clients in the Room)

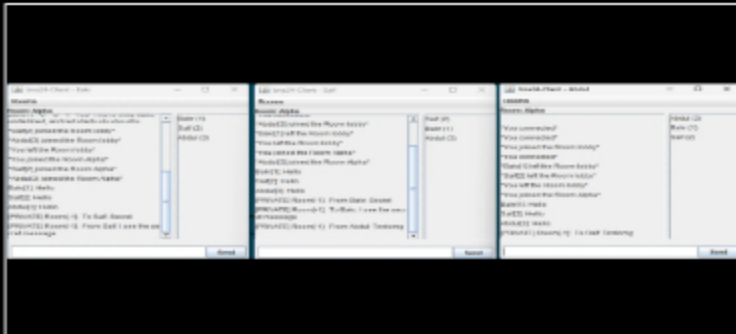
Task Screenshots

Gallery Style: 2 Columns

4

2

1



Private Message Examples

Caption(s) (required) ✓

Caption Hint: *Describe/highlight what's being shown*

Sub-Task

100%

Group: New Features

Task #1: Private messages via @username

Sub Task #2: Show the client-side code that processes the text per the requirement

Task Screenshots

Gallery Style: 2 Columns

```

125 // Send
126 // Remember to use
127
128 private void processClientCommand(String text) throws IOException {
129     if (text.startsWith("@")) {
130         if (myData.getClientName().length() == 0) {
131             System.out.println("You must be the first via /name command", Color.RED);
132             return;
133         }
134         // Handle the case where the user is the first to send a message
135         // Split the text into parts: @username message
136         String[] parts = text.split(" ");
137         String username = parts[1].trim();
138         String message = parts[2].trim();
139         // Check if the target username is valid
140         boolean isValid = false;
141         for (Client client : knownClients) {
142             if (client.getUsername().equals(username)) {
143                 isValid = true;
144                 break;
145             }
146         }
147         if (!isValid) {
148             System.out.println("User not found", Color.RED);
149             return;
150         }
151         // Find the target client ID
152         int targetClientId = -1;
153         for (Client client : knownClients) {
154             if (client.getUsername().equals(username)) {
155                 targetClientId = client.getId();
156                 break;
157             }
158         }
159         // Create a payload object
160         Payload payload = new Payload();
161         payload.setType(PayloadType.PRIVATE_MESSAGE);
162         payload.setTargetClientId(targetClientId);
163         payload.setMessage(message);
164         // Send the payload to the server
165         send(payload);
166     } else {
167         // Handle the case where the user is not the first to send a message
168         // Split the text into parts: @username message
169         String[] parts = text.split(" ");
170         String username = parts[1].trim();
171         String message = parts[2].trim();
172         // Check if the target username is valid
173         boolean isValid = false;
174         for (Client client : knownClients) {
175             if (client.getUsername().equals(username)) {
176                 isValid = true;
177                 break;
178             }
179         }
180         if (!isValid) {
181             System.out.println("User not found", Color.RED);
182             return;
183         }
184         // Find the target client ID
185         int targetClientId = -1;
186         for (Client client : knownClients) {
187             if (client.getUsername().equals(username)) {
188                 targetClientId = client.getId();
189                 break;
190             }
191         }
192         // Create a payload object
193         Payload payload = new Payload();
194         payload.setType(PayloadType.PRIVATE_MESSAGE);
195         payload.setTargetClientId(targetClientId);
196         payload.setMessage(message);
197         // Send the payload to the server
198         send(payload);
199     }
200 }

```

```

201 // Handle
202 // Remember to use
203
204 private void handlePrivateMessageCommand(String text) {
205     String[] parts = text.split(" ");
206     if (parts.length < 3) {
207         System.out.println("Invalid private message format. Use /username /message", Color.RED);
208         return;
209     }
210     String targetUsername = parts[1].trim();
211     String privateMessage = parts[2].trim();
212     // Find the target client ID
213     int targetClientId = -1;
214     for (Client client : knownClients) {
215         if (client.getUsername().equals(targetUsername)) {
216             targetClientId = client.getId();
217             break;
218         }
219     }
220     if (targetClientId == -1) {
221         System.out.println("User not found", Color.RED);
222         return;
223     }
224     // Create a payload object
225     Payload payload = new Payload();
226     payload.setType(PayloadType.PRIVATE_MESSAGE);
227     payload.setTargetClientId(targetClientId);
228     payload.setMessage(privateMessage);
229     // Send the payload to the server
230     send(payload);
231 }

```

If Text Starts With @

handlePrivateMessage

Caption(s) (required) ✓

Caption Hint: Describe/highlight what's being shown

Task Response Prompt

Explain in concise steps how this logically works

Response:

On the client side, private messaging happens when the input text starts with @, signaling a private message. The system parses the input to extract the target username and message data. It then checks the knownClients to validate the target username and get the associated client ID. If the target client ID is found, a Payload object is created with the PRIVATE_MESSAGE type ENUM, including the target client ID and the message content. This payload is sent to the server using the send() method. If the target username is not found, the client displays an error message to the sender, and no payload is sent to the server.

Sub-Task

100%

Group: New Features

Task #1: Private messages via @username

Sub Task #3: Show the ServerThread code receiving the payload and passing it to Room

Task Screenshots

Gallery Style: 2 Columns

4

2

1

```

1 // Handle
2 // Remember to use
3
4 private void handlePrivateMessageCommand(String text) {
5     String[] parts = text.split(" ");
6     if (parts.length < 3) {
7         System.out.println("Invalid private message format. Use /username /message", Color.RED);
8         return;
9     }
10    String targetUsername = parts[1].trim();
11    String privateMessage = parts[2].trim();
12    // Find the target client ID
13    int targetClientId = -1;
14    for (Client client : knownClients) {
15        if (client.getUsername().equals(targetUsername)) {
16            targetClientId = client.getId();
17            break;
18        }
19    }
20    if (targetClientId == -1) {
21        System.out.println("User not found", Color.RED);
22        return;
23    }
24    // Create a payload object
25    Payload payload = new Payload();
26    payload.setType(PayloadType.PRIVATE_MESSAGE);
27    payload.setTargetClientId(targetClientId);
28    payload.setMessage(privateMessage);
29    // Send the payload to the server
30    send(payload);
31 }

```

processPayload CASE PRIVATE_MESSAGE

Caption(s) (required) ✓

Caption Hint: Describe/highlight what's being shown

Task Response Prompt

Explain in concise steps how this logically works

Response:

When the server receives a `PRIVATE_MESSAGE` payload from a client, it first identifies the type of message through the `processPayload` method in the `ServerThread` class. If the payload type matches `PRIVATE_MESSAGE`, the server extracts the target client ID and the message content from the payload. It then uses the current room (`currentRoom`) to call the `sendPrivateMessage` method, passing along the sender's information, the target client ID, and the message. This ensures that the room handles the logic of delivering the message only to the intended recipient and the sender.

Sub-Task

Group: New Features

100%

Task #1: Private messages via @username

Sub Task #4: Show the Room code that verifies the id and sends the message to both the sender and receiver

Task Screenshots

Gallery Style: 2 Columns

4

2

1

```
1 // Room
2 // Message: 01, 000
3 // Payload: {type: 'PRIVATE_MESSAGE', sender: 'Alice', targetClientId: 'Bob', message: 'Hello Bob'}
4 // Action: Send private message to Bob
5 // Result: Success
6 // Error: null
7 // Status: 200
8 // Headers: {}
9 // Body: {}
10 // Content-Type: application/json
11 // Charset: UTF-8
12 // Accept: */*
13 // Accept-Encoding: gzip, deflate
14 // User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/91.0.4472.114 Safari/537.36
15 // Referer: http://localhost:3000/
16 // Host: localhost:3000
17 // Connection: keep-alive
18 // Cache-Control: no-cache
19 // Pragma: no-cache
20 // Expires: 0
21 // Server: Express/4.17.1
22 // Date: Mon, 14 Jun 2021 12:00:00 GMT
23 // Content-Length: 100
24 // ETag: W/"64-1234567890"
25 // Vary: Accept-Encoding
26 // X-Powered-By: Express
27 // X-Request-Id: 1234567890
28 // X-Response-Time: 10ms
29 // X-Content-Type-Options: nosniff
30 // X-Frame-Options: sameorigin
31 // X-XSS-Protection: 1; mode=block
32 // X-WebKit-CSP: default-src 'self'; script-src 'self'; style-src 'self'; font-src 'self'; img-src 'self'; media-src 'self';
33 // X-Permitted-Cross-Domain-Policies: none
34 // X-Debug-Header: {}
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100 // X-Debug-Header: {}
```

sendPrivateMessage

Caption(s) (required) ✓

Caption Hint: Describe/highlight what's being shown

Task Response Prompt

Explain in concise steps how this logically works

Response:

The `sendPrivateMessage` method in the `Room` class handles private messaging by first checking if the room is active and retrieving the target client using their ID from the list of connected clients. If the target client exists, the server formats two distinct messages: one for the sender, indicating the message was sent, and one for the receiver, showing the sender's message. Both messages are then sent using the `sendMessage` method.

End of Task 1

Task

100%

Group: New Features

Task #2: Mute and Unmute

Weight: ~50%

Points: ~2.00

⬆️ COLLAPSE ⬆️

Details:

All code screenshots must include ucid/date.

App screenshots must have the UCID in the title bar like the lesson gave.



- Client-side will implement a /mute and /unmute command (i.e., /mute Bob or /unmute Bob)

Columns: 1

Sub-Task

Group: New Features

Task #2: Mute and Unmute

Sub Task #1: Show a few examples across different clients (there should be at least 3 clients in the Room)

100%

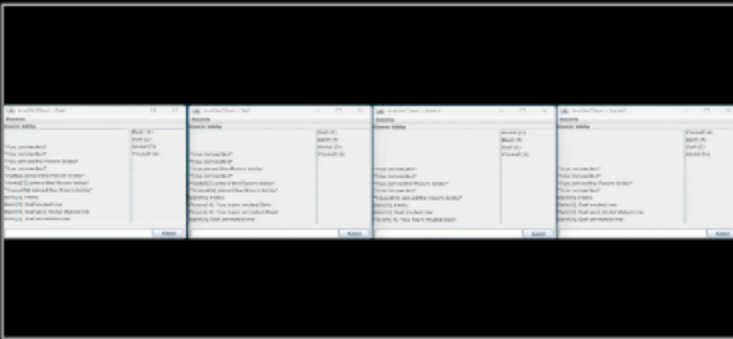
Task Screenshots

Gallery Style: 2 Columns

4

2

1



Mute and Unmute Examples

Caption(s) (required) ✓

Caption Hint: *Describe/highlight what's being shown*

Sub-Task

Group: New Features

Task #2: Mute and Unmute

Sub Task #2: Show the client-side code that processes the text per the requirement

100%

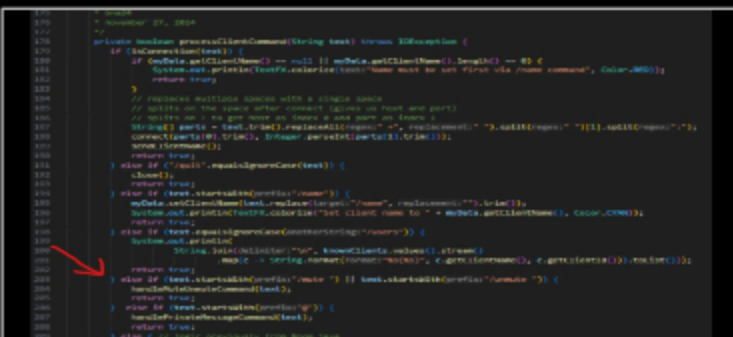
Task Screenshots

Gallery Style: 2 Columns

4

2

1



processClientCommand



handleMuteUnmuteCommand

Caption(s) (required) ✓

Caption Hint: *Describe/highlight what's being shown*

– Task Response Prompt

Task Response Prompt

Explain in concise steps how this logically works

Response:

The first step with the mute/unmute feature (client side) is to identify whether it starts with /mute or /unmute. Once our command is recognized, we pull the username from the command by just stripping off the /mute or /unmute prefix. Then it uses the knownClients map to locate the user ID associated with the specified username. An error message is shown to the client if the user is not found. If user is found, we create a Payload object with a type of either MUTE or UNMUTE and append the target user's ID to payload. The send() method is used to send the payload to the server. Then the client displays a confirmation message containing the result of the specified user having been successfully muted or unmuted.

Sub-Task

Group: New Features

Task #2: Mute and Unmute

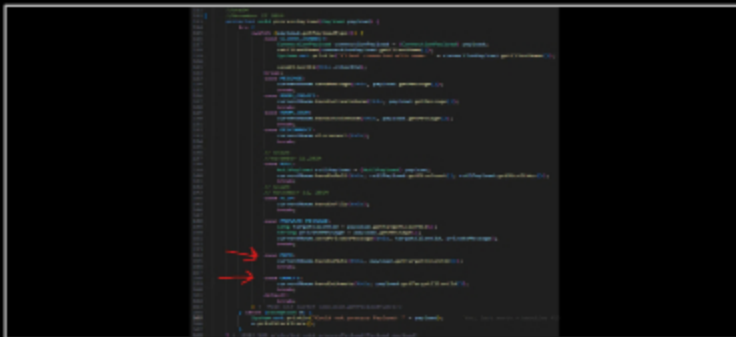
Sub Task #3: Show the ServerThread code receiving the payload and passing it to Room

100%

Task Screenshots

Gallery Style: 2 Columns

4 2 1



processPayload

Caption(s) (required) ✓

Caption Hint: Describe/highlight what's being shown

Task Response Prompt

Explain in concise steps how this logically works

Response:

When a client sends a mute or unmute request, the ServerThread processes it through the processPayload method. This method identifies the type of payload (either MUTE or UNMUTE) and forwards the request to the appropriate handler (handleMute or handleUnmute). These handlers extract the target client's ID from the payload and pass it along with the requesting client to the Room.

Sub-Task

Group: New Features

Task #2: Mute and Unmute

Sub Task #4: Show the Room code that verifies the id and add/removes the muted name to/from the ServerThread's list

100%

Task Screenshots

Gallery Style: 2 Columns

In the Room class, both public and private messages respect the mute list. For public messages, the sendMessage method checks each recipient's mute list and skips delivering the message if the sender is muted, logging the event. For private messages, the sendPrivateMessage method verifies whether the recipient has muted the sender. If so, the message is not delivered, and the sender is informed that their private message was not delivered due to being muted.

Sub-Task

Group: New Features

Task #2: Mute and Unmute

Sub Task #6: Show terminal supplemental evidence per the requirements (refer to the details of this task)

100%

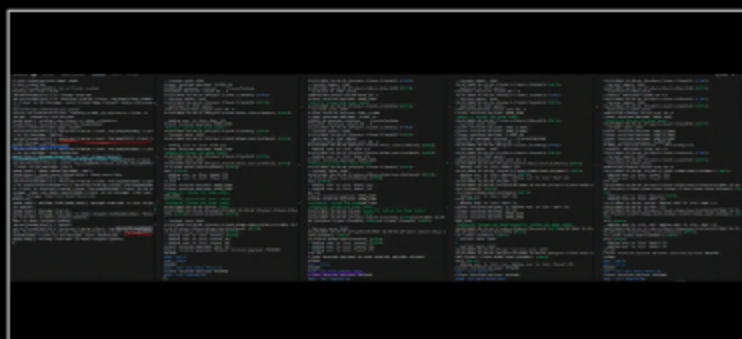
Task Screenshots

Gallery Style: 2 Columns

4

2

1



terminal supplemental evidence

Caption(s) (required) ✓

Caption Hint: *Describe/highlight what's being shown*

End of Task 2

End of Group: New Features

Task Status: 2/2

Group

100%

Group: Misc

Tasks: 3

Points: 1

^ COLLAPSE ^

Task

100%

Group: Misc

Task #1: Add the pull request link for the branch

Weight: ~33%

Points: ~0.33

^ COLLAPSE ^

Details:

Note: the link should end with /pull/#



Task URLs

URL #1

<https://github.com/BakrAbushaar/bna24-IT114-0>

URL

<https://github.com/BakrAbushaar/bna24-IT114-0>

End of Task 1

Task



Group: Misc

Task #2: Talk about any issues or learnings during this assignment

Weight: ~33%

Points: ~0.33

^ COLLAPSE ^

Task Response Prompt

Response:

I had many issues getting the basic UI set up as the chatPanel wouldnt work. I asked for help and the professor guided me through it by telling me what to look for. I got the basics working while getting a deeper understanding of how the project works.

End of Task 2

Task



Group: Misc

Task #3: WakaTime Screenshot

Weight: ~33%

Points: ~0.33

^ COLLAPSE ^

Details:

Grab a snippet showing the approximate time involved that clearly shows your repository. The duration isn't considered for grading, but there should be some time involved



Task Screenshots

Gallery Style: 2 Columns

4

2

1

Files		Branches	
1 hr 30 mins	Server/room.java	4 hrs 20 mins	Milestones
1 hr 21 mins	Client/Client.java		
30 mins	Server/ServerThread.java		
24 mins	Client/Views/ChatPanel.java		
9 mins	Client/Views/ConnectionPanel.java		
7 mins	Client/Views/RoomPanel.java		
7 mins	Client/Views/UserDetailsPanel.java		
4 mins	Client/ClientUI.java		
3 mins	Client/Views/SignUpPanel.java		
34 secs	Common/PayloadType.java		
35 secs	Common/Payload.java		
20 secs	Client/ContentView.java		
15 secs	Client/Interfaces/IClientEvents.java		
12 mins	Common/ToolFX.java		
9 secs	Source.txt		
6 secs	Common/LoggerUtil.java		
5 secs	Client/Interfaces/RoomEvents.java		
4 secs	Client/Views/RoomListItem.java		
3 secs	Client/Interfaces/ConnectionEvents.java		
2 mins	Client/Interfaces/MessageEvents.java		
1 sec	Client/Views/SignUpItem.java		
0 secs	Server/ServerThread.class		
0 secs	Server/Server.class		
0 secs	Server/ServerThread\$1.class		

waketime

End of Task 3

End of Group: Misc
Task Status: 3/3

End of Assignment